Tutorial 2

Ex. 8.2

Q18 $\int \underbrace{(r^2+r+1)}_{\text{diff}} \underbrace{e^r}_{\text{int}} \, dr$

$= (r^2+r+1)e^r$

$- (2r+1)e^r$

$+ 2e^r + C$

Tabular integration

	diff	int
+	r^2+r+1	e^r
−	$2r+1$	e^r
+	2	e^r
−	0	e^r

Q 24

$$\int e^{-2x} \sin 2x \, dx$$

$u = 2x$

$du = 2dx$

$$= \frac{1}{2} \int \underbrace{e^{-u}}_{\text{int}} \underbrace{\sin u}_{\text{diff}} \, du$$

	diff	int
+	$\sin u$	e^{-u}
−	$\cos u$	$-e^{-u}$
+	$-\sin u$	e^{-u}

$$\boxed{\int (-\sin u) e^{-u} \, du}$$

$$= \frac{1}{2} \left(-e^{-u} \sin u + e^{-u} \cos u - \int (\sin u) e^{-u} \, du \right)$$

$$I := \int e^{-u} \sin u \, du$$

$$\Rightarrow I = \frac{1}{2} \left(-e^{-u} \sin u + e^{-u} \cos u - I \right)$$

$$\Rightarrow I = \ldots$$

Q42 $\displaystyle\int \underbrace{\sin 2x}_{\text{diff}} \underbrace{\cos 4x}_{\text{int}}\, dx$

	diff	int
$+$	$\sin 2x$	$\cos 4x$
$-$	$2\cos 2x$	$\frac{1}{4}\sin 4x$
$+$	$-4\sin 2x$	$-\frac{1}{16}\cos 4x$

① Int. by part

$$= \frac{1}{4}\sin 2x \sin 4x + \frac{1}{8}\cos 2x \cos 4x + \int \frac{1}{4}\sin 2x \cos 4x\, dx$$

$$I := \int \sin 2x \cos 4x\, dx \Rightarrow I = \cdots + \frac{1}{4} I$$

$$I = \cdots$$

② Product to Sum

$$\sin A \cos B = \frac{1}{2}\sin(A-B) + \frac{1}{2}\sin(A+B)$$

Q53 $\int (1+2x^2)e^{x^2}\,dx$

$$= \underline{\int 1\cdot e^{x^2}\,dx} + \int 2x^2 e^{x^2}\,dx$$

	diff	int
$+$	e^{x^2}	1
$-$	$2xe^{x^2}$	x

$$= xe^{x^2} - \underline{\int 2xe^{x^2}\cdot x\,dx} + \underline{\int 2x^2 e^{x^2}\,dx}$$

$$= xe^{x^2} + C$$

Ex 8.3

Q24 $\int_0^{\pi} \sqrt{1-\cos 2x}\, dx$

① For $1 \pm \cos 2x$

$1-\cos 2x = 2\sin^2 x$

$= \int_0^{\pi} \sqrt{2\sin^2 x}\, dx \longrightarrow \sqrt{f(x)^2} = |f(x)| \neq f(x)$

$= \int_0^{\pi} \sqrt{2}\sin x\, dx$ or $\underbrace{\int_0^{\pi} \sqrt{2}(-\sin x)\, dx}_{\text{Wrong}}$?

$\sin^2\theta + \cos^2\theta = 1$

$\sqrt{\sin^2 2x} = \sin 2x$

② Mult. by conjugate

$$\int_0^{\pi} \sqrt{1-\cos 2x}\, dx \cdot \frac{\sqrt{1+\cos 2x}}{\sqrt{1+\cos 2x}} = \int_0^{\pi} \frac{\boxed{\sqrt{1-\cos^2 2x}}}{\sqrt{1+\cos 2x}}\, dx$$

$u = 1+\cos 2x$

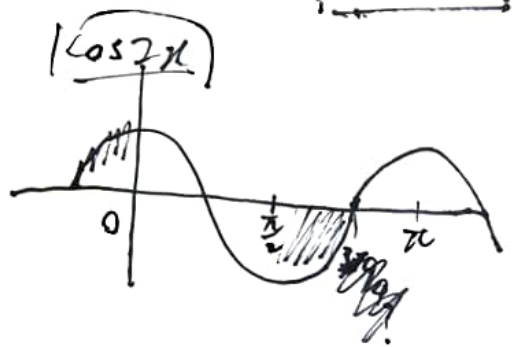
Q30 $\displaystyle\int_{\pi/2}^{3\pi/4} \sqrt{1-\sin 2x}\, dx \cdot \frac{\sqrt{1+\sin 2x}}{\sqrt{1+\sin 2x}}$ $\quad 1 \pm \cos 2\theta$

$\sin 2x = 2\sin x \cos x$

$$= \int_{\pi/2}^{3\pi/4} \frac{\sqrt{1-\sin^2 2x}}{\sqrt{1+\sin 2x}}\, dx$$

$\sqrt{\cos^2 2x} = |\cos 2x|$

$\cos 2x$ $\quad \boxed{-\cos 2x}$



Q48 $\int \cot^6 2x \, dx$ (take out $\cot^2 2x$)

apply $\underline{\cot^2 2x} = \csc^2 2x - 1$

$= \int (\cot^4 2x)(\csc^2 2x - 1) \, dx$

$= \int \underbrace{\cot^4 2x \csc^2 2x}_{u = \cot 2x} \, dx - \int \cot^4 2x \, dx$

$\downarrow$

$\cot^2 2x \cdot \cot^2 2x$

$\downarrow$

$\cot^2 2x (\csc^2 2x - 1)$

Q62 $\int \sin^3\theta \cos 2\theta \, d\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta$

$= \int \sin^3\theta (\cos^2\theta - \sin^2\theta) \, d\theta$

$= \int \sin^3\theta \cos^2\theta \, d\theta - \int \sin^5\theta \, d\theta$

Q64 $\int \sin^1\theta \sin^1 2\theta \sin^1 3\theta \, d\theta$

$\left[\frac{1}{2}\cos(\quad) - \frac{1}{2}\cos(\quad)\right]\sin 3\theta$

Power 1

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Product to Sum

Ex 8.4

$\sqrt{1-x^2} \to x = \sin\theta$ $\quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$1+x^2 \to x = \tan\theta$ $\quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$\sqrt{x^2-1} \to x = \sec\theta$

$\sqrt{\tan^2\theta} = \tan\theta$

$0 \leq \theta < \frac{\pi}{2}$

$\boxed{-\tan\theta}$

We don't want it

Q48 $\int \frac{\sqrt{x-2}}{\sqrt{x-1}} \, dx$

$u = x-1$

$= \int \frac{\sqrt{u-1}}{\sqrt{u}} \, du$

$t = \sqrt{u} \Rightarrow u-1 = t^2-1$

$dt = \frac{1}{2\sqrt{u}} \, du$

$= 2\int \sqrt{t^2-1} \, dt$

Alternative: $w = \sqrt{\dfrac{x-2}{x-1}}$

$$w^2 = \frac{x-2}{x-1}$$

$$w^2 x - w^2 = x - 2$$

$$(w^2 - 1)x = w^2 - 2$$

$$x = \frac{w^2-2}{w^2-1}$$

$$dx = \underline{\left(\qquad\qquad\right)} \, dw$$

$$\int \sqrt{\frac{x-2}{x-1}}\, dx = \int \underline{w\left(\qquad\qquad\right)} \, dw$$